

Unit Rates with Fractions

Finding a unit rate by dividing one fraction by another, comparing two rates by their unit rates, and pulling a unit rate out of a word problem

How to work these problems.

- A **unit rate** tells you how much there is for exactly *one* of something — one hour, one pound, one batch.
- Write the rate as a division: the amount divided by how many units it took. Then divide by multiplying by the reciprocal.
- Turn every mixed number into an improper fraction *before* you divide.
- Give each answer as a fraction in lowest terms (or a whole number when it works out to one), and say what it counts per one unit.

1. Zara jogs $\frac{1}{2}$ of a mile in $\frac{1}{4}$ of an hour. How many miles does she jog per hour?

Answer: _____

2. A printer prints $\frac{3}{4}$ of a page in $\frac{1}{2}$ of a minute. How many pages does it print per minute?

Answer: _____

3. A tap fills $\frac{2}{3}$ of a litre in $\frac{1}{6}$ of a minute. How many litres does it fill per minute?

Answer: _____

4. A recipe uses $1\frac{1}{2}$ cups of flour to make $\frac{3}{4}$ of a batch. How many cups of flour does one whole batch need?

Answer: _____

5. Rowan walks $\frac{3}{5}$ of a kilometre in $\frac{1}{4}$ of an hour. Sadie walks $\frac{2}{3}$ of a kilometre in $\frac{1}{3}$ of an hour. Find each walker's speed in kilometres per hour, then write $>$ or $<$ in the last blank so that "Rowan's speed _____ Sadie's speed" is true.

Rowan: _____ Sadie: _____ Rowan _____ Sadie

6. A tortoise covers $\frac{5}{8}$ of a mile in $\frac{5}{6}$ of an hour. How many miles does it cover per hour?

Answer: _____

7. A hose fills $\frac{7}{8}$ of a barrel in $\frac{1}{2}$ of an hour. How much of a barrel does it fill per hour?

Answer: _____

8. At a market, $\frac{3}{4}$ of a pound of cheese costs 6 dollars. What is the cost of one pound, in dollars?

Answer: _____

9. Painting $\frac{2}{5}$ of a wall takes $\frac{3}{4}$ of an hour. What fraction of the wall gets painted in one hour?

Answer: _____

10. Brand A sells $\frac{2}{3}$ of a pound of trail mix for 2 dollars. Brand B sells $\frac{3}{4}$ of a pound for 3 dollars. Find each brand's price per pound in dollars, then write $>$ or $<$ in the last blank so that "Brand A's price per pound _____ Brand B's price per pound" is true.

Brand A: _____ Brand B: _____ A _____ B

11. A cyclist rides $4\frac{1}{2}$ miles in $\frac{3}{4}$ of an hour. How many miles does the cyclist ride per hour?

Answer: _____

12. A bottling machine fills $2\frac{1}{4}$ litres in $\frac{1}{3}$ of a minute. Working at that same rate, how many litres does it fill in 4 minutes?

Answer: _____